

Cover letter — Journal of Cheminformatics

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[AUTHOR: date]

To the Editors *Journal of Cheminformatics*

Dear Editors,

Please consider the enclosed manuscript, **“No Single Architecture Wins: Ensembles and Conventional Machine Learning Match Pretrained Molecular Models Across 44 Property-Prediction Benchmarks,”** for publication as a **Software article** in the *Journal of Cheminformatics*.

What the paper reports. We present QSARena, an open-source QSAR/AutoML workspace that predicts molecular properties from SMILES through one leakage-controlled workflow, delivered both as a code-free notebook that runs in Google Colab without any installation and as a command-line runner sharing the same core, and we use it to run what is, to our knowledge, the broadest single-tool cross-suite benchmark published to date: 44 datasets from five collections (Therapeutics Data Commons, MoleculeNet, Polaris ADME, PODUAM and ChemML), 28 models and 837 valid model–dataset evaluations, all under one fixed configuration, executed on an NSF ACCESS Jetstream2 A100 node. The model library spans conventional machine learning and gradient boosting, deep tabular and message-passing graph networks, 3D pretrained models (Uni-Mol V1 and V2), MapLight-style descriptor–graph hybrids, and combinatorial-fusion and stacking ensembles.

Why it fits this journal. Three of the paper’s contributions speak directly to debates the *Journal of Cheminformatics* has hosted and shaped:

1. **A unifying benchmarking framework, and an architecture-agnostic finding.** We frame this as a framework-and-finding contribution rather than an accuracy claim, and we are explicit about that in the paper. The framework evaluates 28 models across seven architecture families and five benchmark suites under one leakage-controlled pipeline; the finding is that no family won more than 36% of datasets, which substantially widens the evidence base for something your journal has published repeatedly — that representation and model-family choice, not architectural scale, drive ADMET performance. For context, the best QSARena model per dataset reached the estimated published top ten on 35 of 37 leaderboard-comparable datasets, including all 22 TDC datasets under official `admet_group` splits.

We also state plainly where we do **not** lead. On accuracy QSARena is mid-pack: ADMET-AI holds the highest average TDC rank, MaxQsaring reports first place on 19 of 22 tasks, and Schrödinger’s DeepAutoQSAR is described as a top performer on 20 of 22. A new landscape table (Table 7) positions the tool against fourteen open, commercial and web-based comparators across licence, cost, code-free access, retrainability, architecture breadth, suites benchmarked and published TDC performance — conceding the axes on which competitors lead. We think this is more useful to your readers, and more credible, than a bare novelty assertion.

2. **We quantify the cost of optimistic model selection.** Choosing the best of 28 models using held-out scores — the protocol implicit in most leaderboard submissions — gives the 35-of-37 placement above. Choosing the model by cross-validation alone, with the test set genuinely untouched, drops that to 25 of 37, the median estimated rank from 3 to 8, and first places from five to none; on the official TDC subset it falls from 22 of 22 to 15 of 22. We report both and argue the second is the honest number. We are not aware of another benchmark study that isolates this effect on its own results, and we think it accounts for a meaningful share of the apparent distance between competing leaderboard entries.
3. **Near-state-of-the-art performance with no tuning, no installation and no hardware.** Every result was produced under a *single fixed configuration*, applied unchanged across all 44 datasets, five suites, two task types and a 48-fold range of dataset size, with per-model evolutionary search disabled. There was no per-dataset feature engineering, architecture choice or hyperparameter tuning. What a practitioner gets from one run is therefore what we report — not the endpoint of an expert tuning campaign. That run need not happen on their own machine: the notebook entry point is a single self-contained file that opens in Google Colab, installs its own dependencies in the hosted runtime, and is driven entirely through form widgets, so a user with a CSV of SMILES and measured values writes no code and needs no local CPU, GPU or Python environment. For users who prefer a local or scripted workflow, the same workflow core installs with `pip install qsarena` and exposes a command-line entry point, and is equally suited to batch and HPC use. We also verified this empirically: rerunning the benchmark on a consumer laptop GPU moved the best score by a median of -0.2% across the 37 datasets with byte-identical splits, so the accuracy does not depend on datacentre hardware. For the regulatory and small-laboratory settings your readership includes, where the binding constraint is usually expertise and infrastructure rather than algorithmic novelty, we think this combination — competitive accuracy, obtained without iteration, from a browser — is as substantive a contribution as the benchmark itself.
4. **Reproducibility is built in, not asserted.** Every figure, table and quoted number regenerates from the deposited artifacts with one command (`render_manuscript_assets.py`), and a companion script (`verify_manuscript_numbers.py`) asserts that the manuscript text still matches those artifacts — 64 automated checks that fail loudly if the text and the data drift apart. This directly answers the code-availability and archival criteria set out in your editorial on improving reproducibility and reusability.

We have been deliberately conservative about what we claim. The benchmark is single-split and single-seed: replicating it across five seeds was beyond our compute budget, since the single-seed run alone consumed 112 hours of A100 time. We state this as the principal limitation, spell out its four specific consequences for our tables, and restrict our conclusions to findings that rest on categorical rather than close differences. We also report backend failures rather than analysing only successes, flag that two comparisons are not leaderboard-equivalent because they use locally generated splits, and disclose that our ensemble member filtering consults held-out data. We would rather present a limited result honestly than a stronger one we cannot support.

On naming and prior art. An earlier draft of this work used the name *AutoQSAR*, which collides with an established Schrödinger product of the same name in the same application area. We have renamed the tool **QSARena** (verified free on PyPI and GitHub), added an explicit disambiguation in the Background, and added a new subsection comparing QSARena against both Schrödinger’s AutoQSAR/DeepAutoQSAR and the earlier QSAR Workbench across benchmark breadth, reported performance, licensing, cost and reproducibility. We were careful not to overclaim

there: Schrödinger’s DeepAutoQSAR benchmark reports qualitative criteria without per-dataset metrics or released predictions, so we state plainly that a direct accuracy comparison is not possible and restrict the comparison to the axes that can be evidenced.

Declarations. This manuscript is original, is not under consideration elsewhere, and has not been published previously. The author declares no competing interests. The work was supported by the California Office of Environmental Health Hazard Assessment. Computation used NSF ACCESS Jetstream2 under allocation CIS261142. All software and benchmark artifacts are openly available under the MIT License at <https://github.com/ScottCoffin/QSARena>, with a public issue tracker, and a versioned Zenodo archive will be deposited before publication. [AUTHOR: if you post a preprint, add here — “A preprint of this manuscript has been deposited at arXiv (DOI/ID: ...) under a CC BY license.”]

Suggested reviewers. [AUTHOR: optional — the journal invites 3–5 suggestions. Candidates whose work this paper engages directly include authors of the TDC reproducibility assessment, the MapLight submission, DeepMol, Auto-ADMET and the feature-representation benchmarking literature. Please confirm none has a conflict of interest with you.]

Thank you for considering this work.

Sincerely,

Scott Coffin California Office of Environmental Health Hazard Assessment